



EDUCATION & ACHIEVEMENTS			
Program	Institute	% / CGPA	Year
M.Tech. (Data Science & Artificial Intelligence)	IIT Madras	8.82	2027
B.Tech. (Computer Science)	SGSITS Indore	6.54	2023
Class XII (CBSE)	Gyan Ganga Academy, Bhopal	88.0 %	2019
Class X (CBSE)	M.G.M. Higher Secondary School, Itarsi	9.40	2017
• Secured AIR 35 in GATE 2025 (Data Science & AI) while working full-time as a Junior Engineer at Indian Railways.			
PROFESSIONAL EXPERIENCE			
Native AI Intern Wipro Ltd.		June 2026 – July 2026 Chennai	
• Developed <code>ssctl</code> , a governed CLI interface between AI assistants and SkillSpaces, enabling standardized SkillFlow discovery, execution, and controlled workflow access.			
• Built a React/TypeScript UI for real-time SkillFlow visualization, using Server-Sent Events (SSE) to track workflow execution and support human approval and handoff gates.			
• Implemented artifact management and structured skill execution, using Pydantic validation and feedback-driven retries to reliably pass outputs between downstream workflow nodes.			
RESEARCH EXPERIENCE			
GeoCVS: Auditable Geometric Assessment of Surgical Safety (Master's Thesis)		Aug 2026 – Present	
Thesis Advisor: Dr. Vinkle Srivastav			
• Investigating an interpretable alternative to black-box AI models for surgical safety assessment (Critical View of Safety in laparoscopic cholecystectomy), replacing learned safety scores with explicit, contestable geometric measurements.			
• Designing a pipeline combining few-shot segmentation via SAM 3 with monocular depth estimation (Depth Anything 3) and conformal-prediction-based uncertainty calibration, targeting the SAGES CVS Challenge benchmark.			
TECHNICAL PROJECTS			
Driver Drowsiness Detection System (Course Project: DA5402 – ML Operations Lab)			
• Developed a computer vision system to classify drivers as alert or drowsy, training XGBoost and SVM on 27,000+ feature windows to reach a peak F1 of 0.934 (AUROC 0.994).			
• Engineered 9 clinical drowsiness features (EAR, PERCLOS, head pose) from MediaPipe landmarks via PySpark, versioned through a 7-stage DVC pipeline orchestrated by 2 Airflow DAGs.			
• Built a closed-loop MLOps system where real-time KL-divergence drift detection (Prometheus/Grafana) triggers Airflow retraining and promotes models to the MLflow registry only after a >2% F1 gain.			
Neural Networks & Transformers from Scratch (Course Project: DA6401 – Introduction to Deep Learning)			
• Implemented a fully-connected neural network from scratch in NumPy with forward/backpropagation, six optimizers (SGD, Momentum, NAG, RMSProp, Adam, Nadam), and Xavier initialization, achieving 96.4% test accuracy on MNIST.			
• Implemented the Transformer architecture from scratch in PyTorch (multi-head attention, positional encoding) for German→English translation on Multi30k, reaching a BLEU score of 39.4 with beam-search decoding.			
Panoramic Image Stitching (Course Project: EE5178 – Modern Computer Vision)			
• Built an image-stitching pipeline entirely from scratch (SIFT matching, DLT homography via SVD, and a custom RANSAC) with inverse-warped, alpha-blended compositing across multi-image sequences.			
• Diagnosed ghosting artifacts in near-field handheld shots as a motion-parallax violation of the homography's planar and pure-rotation assumption, and proposed a tripod-based fix rotating about the camera's optical center.			
SKILLS			
Languages: Python, C++, SQL MLOps & Data Engineering: DVC, Apache Airflow, MLflow, Docker, Prometheus, Grafana AI/ML & Deep Learning: PyTorch, NumPy, Pandas, Scikit-learn Tools: Git, FastAPI, W&B, Linux, Jupyter, PySpark, VS Code			
RELEVANT COURSEWORK			
Foundations of Machine Learning	Artificial Intelligence	Data Structures & Algorithms	Data Analytics Lab
Introduction to Deep Learning	Modern Computer Vision	Natural Language Processing	ML Operations Lab
POSITIONS OF RESPONSIBILITY			
• Placement Coordinator, Data Science & AI: Coordinated recruiter interactions and interview scheduling for the cohort.			
• Teaching Assistant for Linear Algebra & Data Structures for Data Science: Conducted tutorials and graded assignments.			